

# Constrained Torque-Level Formation Control of Differential-Drive Wheeled Mobile Robots via Consensus-Based Feedback Linearised MPC

Faisal Lawan, Weishu Zhan, Wei Pan, Joaquin Carrasco

**Abstract**—This paper presents a distributed, torque-level formation control framework for differential-drive wheeled mobile robots. While nonlinear MPC (NMPC) and feedback-linearised MPC methods can achieve accurate formation tracking, their torque-level implementations are hindered by non-convex constraints, which lead to high computational cost. We address this challenge by combining feedback linearisation with a distributed MPC architecture and introducing a tractable Second-Order Cone Programming (SOCP) reformulation of the resulting non-convex problem. The proposed approach captures actuator-level dynamics while convexifying state-dependent constraints, enabling efficient optimisation with formal guarantees of recursive feasibility and local asymptotic stability. We benchmark the controller against NMPC, Linear MPC, and non-convex feedback-linearised MPC. Simulations show that the SOCP achieves torque smoothness comparable to the non-convex formulation while reducing average solver time from tens of milliseconds to around 2 ms in MATLAB, making it suitable for real-time implementation on computationally constrained platforms. These results position the method as a promising foundation for torque-level, distributed formation control on constrained systems, with validation on a real-world system identified as future work.

## I. INTRODUCTION

The art of coordinating multiple autonomous robots to perform tasks collectively is a cornerstone of modern robotics, with applications spanning logistics [1], surveillance [2], and environmental monitoring [3]. A critical capability for these multi-robot systems (MRS) is formation control: the problem of steering a group of robots to achieve and maintain a desired geometric pattern while navigating through an environment [4]. This challenge is particularly acute for differential-drive wheeled mobile robots (WMRs), whose motion is restricted by non-integrable velocity constraints. Indeed, Brockett’s theorem establishes that such systems cannot be asymptotically stabilised to a fixed configuration using smooth, time-invariant state feedback [5], [6], motivating the need for more sophisticated control strategies. Designing controllers that are decentralised, constraint-aware, and supported by formal guarantees remains an important open problem [7].

Over the past decades, several canonical approaches have been explored, yet each presents limitations. Leader-follower strategies are intuitive to implement but introduce a single point of failure and can amplify errors downstream [4]. Virtual structure methods provide rigid formations but often

rely on global position information, thereby limiting scalability [8]. Meanwhile, behaviour-based approaches provide modularity but typically lack formal stability guarantees [9]. These methods, while foundational, do not simultaneously address nonholonomic constraints, actuator dynamics, and decentralised coordination with rigorous guarantees.

Distributed Model Predictive Control (DMPC) has emerged as a principled alternative, offering constraint-handling and cooperative behaviour through local optimisation [10]. However, solving nonlinear programs online introduces bottlenecks on scalability and real-time implementation [11]. A representative baseline [12] demonstrates this trade-off: nonlinear MPC (NMPC) accurately captures dynamics but requires a non-convex, computationally expensive optimisation; linear MPC (LMPC) is faster but relies on successive Jacobian linearisations, degrading performance when the system deviates from its nominal trajectory. This exposes a clear gap: a need for a control strategy that attains NMPC-level fidelity while preserving the tractability of convex optimisation.

Many differential-drive platforms accept velocity or wheel-speed commands and rely on low-level controllers to realise them, which makes kinematic and velocity-level formation controllers both attractive and, in practice, effective. Their commands are issued in velocity coordinates, however, whereas the hard limits to be respected, motor torque and wheel speed, act at the actuator level and depend on the robot’s orientation. Enforcing those limits with feasibility and stability guarantees therefore calls for reasoning directly in wheel torques rather than delegating them to an inner loop [13]. We therefore address the limitations above with a distributed, torque-level formation control framework that unifies tracking performance and computational tractability, capturing the actuator-level limits that kinematic controllers abstract away. Our approach builds upon a DMPC architecture governed by a consensus protocol [14] and applies a two-stage input-output feedback linearisation to obtain an exact linear double-integrator error model [15]. This contrasts with existing feedback-linearised MPC approaches that either operate at the kinematic level [16], [17] or do not support decentralised, constrained formation control [18].

A central contribution of this work is a tractable reformulation of the inherently non-convex constrained MPC problem. By deriving conservative inner approximations of the state- and input-dependent constraints arising from feedback linearisation (following similar principles to [16], [17]), we recast the problem as a Second-Order Cone Programme (SOCP), thereby enabling computationally efficient analysis

W. Zhan and W. Pan are with the Department of Computer Science, The University of Manchester, M13 9PL, United Kingdom.

F. Lawan and J. Carrasco are with the Department of Electrical and Electronic Engineering, The University of Manchester, M13 9PL, United Kingdom, (email: [faisal.lawan@manchester.ac.uk](mailto:faisal.lawan@manchester.ac.uk))

and feasibility guarantees.

Feedback linearisation, consensus-based MPC, and second-order cone programming are each well understood in isolation. Their value here lies in how they are brought together: actuator dynamics are carried through to the torque level, and the constrained problem that results is convexified so that hard actuator limits are respected at real-time rates with feasibility and stability guarantees intact. Concretely, the contributions are: 1) A torque-level, feedback-linearised DMPC formulation for distributed formation control of differential drive robots, enabling accurate tracking of formation objectives at the actuator level; 2) A tractable SOCP reformulation of the non-convex optimal control problem via inner approximations of state-dependent constraints; 3) Formal guarantees of recursive feasibility and local asymptotic stability, benchmarked in simulation against NMPC, LMPC, and non-convex feedback-linearised MPC (NFLMPC).

## II. PROBLEM FORMULATION

This section formalises the problem of distributed formation control for a team of  $n$  differential drive WMRs.

### A. Notation

$\mathbb{R}$  denotes the set of real numbers, and  $\mathbb{R}^{n \times m}$  denotes the set of  $n \times m$  real matrices. Bold symbols denote vectors or matrices (e.g.,  $\mathbf{x}$ ,  $\mathbf{A}$ ), while scalars are non-bold (e.g.,  $a$ ,  $\alpha$ ).  $\mathbf{I}_n$  denotes the  $n \times n$  identity matrix. The Euclidean norm is  $\|\cdot\|_2$ , and the weighted quadratic norm is  $\|\mathbf{x}\|_Q^2 = \mathbf{x}^\top \mathbf{Q} \mathbf{x}$ . The operator  $\text{diag}(\cdot)$  forms an ordered diagonal matrix from a vector, and for a matrix  $\mathbf{A} \in \mathbb{R}^{n \times n}$ ,  $\sigma_{\max}(\mathbf{A})$  denotes the largest singular value. For vectors  $\mathbf{x} \in \mathbb{R}^n$ ,  $\mathbf{y} \in \mathbb{R}^m$ , let  $(\mathbf{x}, \mathbf{y}) := [\mathbf{x}^\top, \mathbf{y}^\top]^\top$ . For  $d > 0$ , let  $[d] := \{1, 2, \dots, d\}$ . For a discrete-time prediction horizon with sampling period  $\delta_t$ , a variable at prediction step  $l$  from time  $k$  is denoted  $(\cdot)_{l|k}$ .

### B. Preliminaries

**Definition 1 (Graph Theory [14]):** The communication network is a directed graph  $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{A})$ , where  $\mathcal{V} = [n]$  is the set of robots,  $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$  is the set of communication links, and  $\mathcal{A} = [a_{ij}] \in \mathbb{R}^{n \times n}$  is the weighted adjacency matrix. An edge  $(j, i) \in \mathcal{E}$  implies that robot  $i$  receives information from robot  $j$ . The graph Laplacian is  $\mathcal{L} = \mathcal{D} - \mathcal{A}$ , where  $\mathcal{D} = \text{diag}(\mathbf{d})$  is the in-degree matrix and  $d_i = \sum_{j=1}^n a_{ij}$ . A virtual centre (indexed by 0) provides a reference trajectory to a subset of followers, and is defined by the pinning matrix  $\mathcal{B} = \text{diag}(\mathbf{b})$ , where  $b_i > 0$  if robot  $i$  communicates with the virtual centre.

### C. System Dynamics

We consider a team of  $n$  identical differential-drive WMRs, whose dynamics are derived using the Lagrangian formulation [19]. The configuration of robot  $i \in \mathcal{V}$  is denoted by  $\mathbf{q}_i \triangleq (x_{c,i}, y_{c,i}, \theta_i) \in \mathbb{R}^2 \times \Theta$ , where  $(x_{c,i}, y_{c,i})$  the Cartesian position of the point being tracked on the robot, and  $\theta_i$  is the robot's heading, with  $\Theta \triangleq (-\pi, \pi]$ .

The reduced-order wheel dynamics, derived via projecting the Lagrangian system onto the Pfaffian constraint manifold, are:

$$\overline{\mathbf{M}}(\mathbf{q}_i) \ddot{\Phi}_i + \overline{\mathbf{C}}(\mathbf{q}_i, \dot{\mathbf{q}}_i) \dot{\Phi}_i = \overline{\mathbf{B}}_{\text{sys}} \boldsymbol{\tau}_i, \quad (1)$$

where  $\dot{\Phi}_i(t) \in \mathbb{R}^2$  is the wheel velocities vector and  $\boldsymbol{\tau}_i(t) \in \mathbb{R}^2$  is the motor torques vector, which are physically constrained as follows:  $\dot{\Phi}_{\text{lb}} \leq \dot{\Phi}_i \leq \dot{\Phi}_{\text{ub}}$  and  $\boldsymbol{\tau}_{\text{lb}} \leq \boldsymbol{\tau}_i \leq \boldsymbol{\tau}_{\text{ub}}$ . The matrices  $\overline{\mathbf{M}} : \mathbb{R}^3 \rightarrow \mathbb{R}^{2 \times 2}$ ,  $\overline{\mathbf{C}} : \mathbb{R}^3 \times \mathbb{R}^3 \rightarrow \mathbb{R}^{2 \times 2}$ , and  $\overline{\mathbf{B}}_{\text{sys}} \in \mathbb{R}^{2 \times 2}$  represent the modified Inertia, Coriolis, and input mappings, respectively. For identical drive motors, the input mapping is trivially nonsingular,  $\overline{\mathbf{B}}_{\text{sys}} = \mathbf{I}_2$ . Defining the full state  $\mathbf{x}_i = (\mathbf{q}_i, \dot{\Phi}_i) \in \mathbb{X}$ , where  $\mathbb{X} := \{\mathbf{x} \in \mathbb{R}^5 \mid \dot{\Phi}_{\text{lb}} \leq \dot{\Phi} \leq \dot{\Phi}_{\text{ub}}\}$ , the system can be expressed in the following control-affine form:

$$\dot{\mathbf{x}}_i = \mathbf{f}(\mathbf{x}_i) + \mathbf{g}(\mathbf{x}_i) \boldsymbol{\tau}_i, \quad (2)$$

where  $\mathbf{f} : \mathbb{R}^2 \times \Theta \times \mathbb{R}^2 \rightarrow \mathbb{R}^5$ ,  $\mathbf{g} : \mathbb{R}^2 \times \Theta \times \mathbb{R}^2 \rightarrow \mathbb{R}^{5 \times 2}$ ,  $\boldsymbol{\tau}_i \in \mathbb{U}$ , and  $\mathbb{U} := \{\boldsymbol{\tau} \in \mathbb{R}^2 \mid \boldsymbol{\tau}_{\text{lb}} \leq \boldsymbol{\tau} \leq \boldsymbol{\tau}_{\text{ub}}\}$ .

### D. Formation Control Objective

The control objective is to design distributed torque inputs  $\boldsymbol{\tau}_i$  that drive the consensus-based formation error to zero. Let  $\Delta_i(t) \in \mathbb{R}^2$  be the desired displacement of robot  $i$  from the virtual centre's trajectory  $\mathbf{P}_{\text{vc}}(t) \in \mathbb{R}^2$ . Henceforth, we shall drop the arguments of the functions  $\mathbf{P}_{\text{vc}}(t)$  and  $\Delta_i(t)$  for notational simplicity; however, these are still functions of time and shall be treated as such. The consensus-based formation error for robot  $i$  is

$$\mathbf{e}_i = b_i((\mathbf{P}_i - \Delta_i) - \mathbf{P}_{\text{vc}}) + \sum_{j \in \mathcal{N}_i} a_{ij}((\mathbf{P}_i - \Delta_i) - (\mathbf{P}_j - \Delta_j)), \quad (3)$$

where  $\mathcal{N}_i$  is the set of neighbours of robot  $i$ . This rewrites compactly as an affine transformation:

$$\mathbf{e}_i(t) = \mathbf{h}_i(t) + g_i \mathbf{P}_i(t), \quad (4)$$

where  $g_i = d_i + b_i$  is the effective coupling weight for robot  $i$ , and  $\mathbf{h}_i(t) = -b_i(\mathbf{P}_{\text{vc}} + \Delta_i) - \sum_{j \in \mathcal{N}_i} a_{ij}(\mathbf{P}_j - \Delta_j + \Delta_i)$  aggregates the time-varying trajectory signals of neighbours and the virtual centre.

Stacking all robot positions into  $\mathbf{P} = [\mathbf{P}_1, \dots, \mathbf{P}_n] \in \mathbb{R}^{2 \times n}$  and displacements into  $\Delta = [\Delta_1, \dots, \Delta_n] \in \mathbb{R}^{2 \times n}$ , the global formation error of all robots can be written compactly as:

$$\mathbf{e} = (\mathbf{P} - \Delta) \mathcal{H} - \mathbf{P}_{\text{vc}}(\mathbf{I}_n^\top \mathcal{B}), \quad (5)$$

where  $\mathcal{H} = \mathcal{L} + \mathcal{B}$  is the modified graph Laplacian.

### E. Assumptions

**Assumption 1:**  $\mathcal{G}$  contains a directed spanning tree rooted at the virtual centre, ensuring  $\mathcal{H}$  is nonsingular.

**Assumption 2:** Communication is static, lossless, and instantaneous; states  $\mathbf{x}_i$  are perfectly observable.

**Assumption 3:** Trajectories  $t \mapsto (\mathbf{P}_{\text{vc}}, \dot{\mathbf{P}}_{\text{vc}}, \ddot{\mathbf{P}}_{\text{vc}})$  and  $t \mapsto (\Delta_i, \dot{\Delta}_i, \ddot{\Delta}_i)$  are predefined and bounded. In addition, the virtual centre's heading signal  $\alpha_{\text{vc}}(t) \in \Theta$  and its derivative  $\dot{\alpha}_{\text{vc}}(t) \in \mathbb{R}$  are assumed to be predefined and bounded.

*Assumption 4:* The robot's geometric parameters  $D$ ,  $b$ , and  $r$  are positive. (See Fig. 1)

*Assumption 5:* Over the finite prediction horizon ( $l > 0$ ), each agent approximates the predicted neighbour velocities using a zero-order hold:

$$\dot{\mathbf{P}}_j(k+l|k) = \dot{\mathbf{P}}_j(k), \quad l = 1, \dots, N-1.$$

*Remark 1:* Assumption 5 decouples the distributed MPC: fixing the *predicted* neighbour velocities to the values observed at step  $k$  lets each agent solve its OCP independently. Since predictions are corrected at every sampling instant, this locally constant-velocity approximation stays accurate enough for short-horizon tractability without degrading coordination.

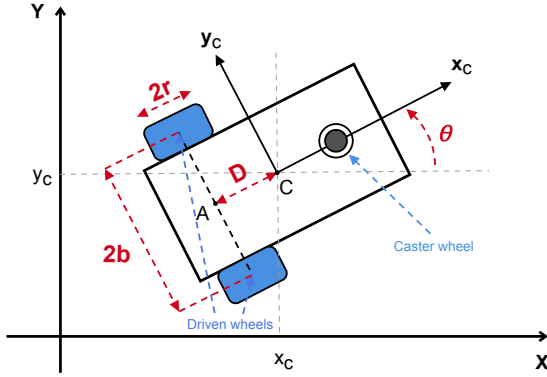


Fig. 1. Robot Model. The tracked point  $[x_c, y_c]^T$  is offset by a distance  $D$  from the wheel axle centre.  $b$  is half of the wheel axle's length, and the wheel radius is  $r$ .

### III. CONTROLLER DESIGN

Our architecture has two components: a feedback linearisation layer that algebraically transforms the complex nonlinear robot dynamics into a simpler, linear representation, and a high-level distributed NMPC that computes optimal torque commands based on this linearised model.

#### A. Feedback Linearisation

The nonlinear mapping from the motor torques  $\tau_i$  to the robot's Cartesian position  $\mathbf{P}_i$  has a relative degree of two. For simplicity and interpretability, we employ a two-stage input-output feedback linearisation [6], [15].

**Stage 1: Actuator Linearisation.** We select the physical torque input to cancel state-dependent nonlinearities in (1):

$$\tau_i = \mathbf{B}_{\text{sys}}^{-1} \left( \overline{\mathbf{M}}(\mathbf{q}_i) \nu_i + \overline{\mathbf{C}}(\mathbf{q}_i, \dot{\mathbf{q}}_i) \dot{\Phi}_i \right), \quad (6)$$

yielding the linear relationship  $\ddot{\Phi}_i = \nu_i$ , where  $\nu_i \in \mathbb{R}^2$  can be physically interpreted as the wheel accelerations. Since  $\mathbf{B}_{\text{sys}} = \mathbf{I}_2$  for the considered robot platform, this inversion is trivially well-defined.

**Stage 2: Error Linearisation.** The formation error has a relative degree of two with respect to  $\nu_i$ . Differentiating  $\mathbf{e}_i$  twice yields  $\ddot{\mathbf{e}}_i = \mathbf{f}_e(\mathbf{x}_i, t) + \mathbf{g}_e(\mathbf{x}_i) \nu_i$ , where  $\mathbf{f}_e : \mathbb{X} \times \mathbb{R} \rightarrow \mathbb{R}^2$  is a function of the system state and the reference

trajectories, and  $\mathbf{g}_e : \mathbb{X} \rightarrow \mathbb{R}^{2 \times 2}$  is the decoupling matrix. Selecting the control law

$$\nu_i = \mathbf{g}_e(\mathbf{x}_i)^{-1} (-\mathbf{f}_e(\mathbf{x}_i, t) + \ddot{\xi}_i), \quad (7)$$

achieves a linear, decoupled double-integrator  $\ddot{\mathbf{e}}_i = \ddot{\xi}_i$ , where  $\xi_i \in \mathbb{R}^2$  is the virtual acceleration input. The exactness of this cancellation relies on accurate knowledge of the inertial and geometric parameters in  $\overline{\mathbf{M}}$  and  $\overline{\mathbf{C}}$ , an idealisation that we adopt here and to which we return in Section VII.

*Lemma 1:* Under Assumption 4, the decoupling matrix  $\mathbf{g}_e(\mathbf{x}_i)$  is nonsingular  $\forall \mathbf{x}_i \in \mathbb{X}$ .

*Proof:* See Section VIII-A. ■

Defining the linear error state as  $\eta_i = (\mathbf{e}_i, \dot{\mathbf{e}}_i) \in \mathbb{R}^4$ , the dynamics under zero-order hold (ZOH) discretisation with sampling period  $\delta_t > 0$  become:

$$\eta_{i,k+1} = \hat{\mathbf{H}} \eta_{i,k} + \hat{\mathbf{G}} \xi_{i,k}, \quad (8)$$

where  $\hat{\mathbf{H}} \in \mathbb{R}^{4 \times 4}$  and  $\hat{\mathbf{G}} \in \mathbb{R}^{4 \times 2}$ . This linear model serves as the prediction model for the NMPC.

*Assumption 6:* The sampling period  $\delta_t > 0$  is sufficiently small such that asymptotic stability of the discrete-time closed-loop system under ZOH implies asymptotic stability of the corresponding continuous-time closed-loop system.

*Remark 2:* The CT/DT consistency in Assumption 6 is motivated by the sampled-data results of [20], which give sufficient conditions under which a controller stabilising an approximate discrete-time model also practically stabilises the exact sampled-data model.

#### B. Conditions for Stability and Feasibility

To guarantee recursive feasibility and asymptotic stability, the analysis follows the standard direct method for MPC, which employs the optimal cost as a candidate Lyapunov function [21]. The standard approach is to construct a feasible candidate control sequence at time  $k+1$  from the optimal solution at  $k$ , augmented with a terminal control action  $\kappa_f(\cdot)$  derived from a local stabilising law:

$$\tilde{\xi}_{i,k+1} = \{\xi_{i,1|k}^*, \dots, \xi_{i,N-1|k}^*, \kappa_f(\eta_{i,N|k}^*)\}. \quad (9)$$

Constrained MPC design [21] dictates that we formally construct a terminal constraint set  $\mathcal{X}_f$  and a locally stable terminal controller  $\kappa_f(\eta_i) = -\mathbf{K} \eta_i$ , where  $\mathbf{K} \in \mathbb{R}^{2 \times 4}$ , that satisfy the following conditions by design:

**Condition 1 (Terminal Admissibility):** For any state  $\eta_i \in \mathcal{X}_f$ , the terminal control law yields a state and input pair that satisfies the system constraints:  $(\mathbf{x}_i, \tau_i) \in (\mathbb{X} \times \mathbb{U})$  (i.e., in this case, this is defined by the wheel velocity constraints and the torque constraints).

**Condition 2 (Terminal Invariance):** The terminal set  $\mathcal{X}_f$ , a closed set containing the origin, is positively invariant for the error dynamics under  $\kappa_f$ . That is, for any  $\eta_i \in \mathcal{X}_f$ , the successor state  $\eta_i^+ = \hat{\mathbf{H}} \eta_i + \hat{\mathbf{G}} \kappa_f(\eta_i) \in \mathcal{X}_f$ .

**Condition 3 (Terminal Cost Decrease):** The terminal cost function  $\ell_f(\eta_i)$  is a Control Lyapunov Function (CLF) satisfying the inequality:

$$\ell_f(\hat{\mathbf{H}} \eta_i + \hat{\mathbf{G}} \kappa_f(\eta_i)) - \ell_f(\eta_i) \leq -\ell_s(\eta_i, \kappa_f(\eta_i)). \quad (10)$$

These conditions are met by selecting  $\mathbf{K}$  as the infinite-horizon LQR gain, where  $\mathbf{P}$  is the solution to the corresponding Discrete Algebraic Riccati Equation (DARE).

### C. Distributed NMPC Formulation

At each time step  $k$ , every robot  $i$  solves a finite-horizon optimal control problem (OCP) to find a sequence of control inputs  $\Xi_{i,k}^* = \{\xi_{i,0|k}^*, \dots, \xi_{i,N-1|k}^*\}$  that minimises a cost function over a prediction horizon  $N > 0$ . The cost function to be minimised is given by the following:

$$J_i(\eta_{i,0|k}, \Xi_{i,k}) = \ell_f(\eta_{i,N|k}) + \sum_{l=0}^{N-1} \ell_s(\eta_{i,l|k}, \xi_{i,l|k}). \quad (11)$$

Given the initial error state  $\eta_{i,0|k} = \eta_i(t_k)$ , robot  $i$  solves the following OCP, denoted Problem  $P(i)$ :

$$\Xi_{i,k}^* = \arg \min_{\Xi_{i,k}} J_i(\eta_{i,0|k}, \Xi_{i,k}) \quad (12a)$$

s.t.

$$\eta_{i,l+1|k} = \hat{\mathbf{H}}\eta_{i,l|k} + \hat{\mathbf{G}}\xi_{i,l|k}, \quad \forall l \in [0, N-1], \quad (12b)$$

$$(\mathbf{x}_{i,l|k}, \boldsymbol{\tau}_{i,l|k}) \in (\mathbb{X} \times \mathbb{U}), \quad \forall l \in [0, N-1], \quad (12c)$$

$$\eta_{i,N|k} \in \mathcal{X}_f. \quad (12d)$$

The stage cost  $\ell_s(\eta, \xi) = \|\eta\|_{\mathbf{Q}}^2 + \|\xi\|_{\mathbf{U}}^2$  penalises formation error and control effort, with weighting matrices  $\mathbf{Q} \in \mathbb{R}^{4 \times 4}$ ,  $\mathbf{Q} \succ 0$  and  $\mathbf{U} \in \mathbb{R}^{2 \times 2}$ ,  $\mathbf{U} \succ 0$ . The terminal cost  $\ell_f(\eta) = \|\eta\|_{\mathbf{P}}^2$ , with weighting matrix  $\mathbf{P} \in \mathbb{R}^{4 \times 4}$ ,  $\mathbf{P} \succ 0$ , is a quadratic penalty term, and  $\mathcal{X}_f$  defines the terminal constraint to guarantee local asymptotic stability. Upon solving the OCP, only the first element of the optimal sequence,  $\xi_{i,0|k}^*$ , is implemented and mapped back to the physical torque. The problem is reevaluated at the next time step,  $k+1$ , in a receding horizon fashion.

Notice that although the objective function of Problem  $P(i)$  is convex, the constraints are nonconvex and highly nonlinear. Thus, this is a nonconvex nonlinear programming problem, and in general, it is NP-hard to solve globally. Consequently, there is no known algorithm capable of guaranteeing a global optimum in polynomial time. In the worst case, computation times scale exponentially, rendering the standard formulation intractable for the strict real-time deadlines of mobile robot control. This motivates reformulating the problem as a simpler class while preserving solution quality, as discussed further in Section V.

## IV. TERMINAL CONSTRAINT SET DESIGN

A significant challenge arises because the feedback linearisation transforms the simple, constant bounds on actuator torques and wheel velocities into complex, state-dependent constraints on the synthetic input  $\xi_i$ . To enforce Conditions 1 and 2, we construct the terminal set based on a conservative inner approximation of the true admissible set [17].

Substituting (7) into (6), the exact physical torque decom-

poses as:

$$\boldsymbol{\tau}_i = \underbrace{\mathbf{w}(\eta_i)^\top \mathbf{H}(\theta_i) \mathbf{w}(\eta_i)}_{\boldsymbol{\tau}_{\text{fb},i}} \begin{bmatrix} -1 \\ 1 \end{bmatrix} + \underbrace{c_i \sigma \boldsymbol{\mu}(\theta_i) (\xi_i - \ddot{\mathbf{h}}_i)}_{\boldsymbol{\tau}_{\text{ff},i}}, \quad (13)$$

where  $\mathbf{w}(\eta_i) = \dot{\mathbf{e}}_i(t) - \dot{\mathbf{h}}_i(t) \in \mathbb{R}^2$  represents the modified error rate based on the aggregated neighbour trajectories defined in (4). Here,  $\mathbf{H}(\theta_i) = \mathbf{R}(\theta_i + \pi/4) \cdot \text{diag}(\lambda_i, -\lambda_i) \cdot \mathbf{R}(\theta_i + \pi/4)^\top$ , and  $\boldsymbol{\mu}(\theta_i) \in \mathbb{R}^{2 \times 2}$  is an orientation-dependent map from the synthetic input space to the torque space. The physical constants appearing in the constraint set are derived directly from the robot geometry and graph topology:

$$c_i = \frac{r}{2b} \frac{1}{g_i D}, \quad \lambda_i = \frac{1}{2} \frac{J_{B,C}}{g_i D}, \quad z_i = \frac{\sqrt{b^2 + D^2}}{r g_i D},$$

$$\sigma = \sqrt{(mbD)^2 + (J_B + mD^2)^2},$$

where  $m$  is the robot mass and  $J_B$  is the moment of inertia about the body-frame origin.

To guarantee satisfaction of the physical constraints, expressed as  $\|\boldsymbol{\tau}_i\|_\infty \leq \tau_{\max}$  and  $\|\dot{\Phi}_i\|_\infty \leq \Phi_{\max}$ , for all robot orientations  $\theta_i$ , we derive sufficient,  $\theta$ -independent conditions. For the torque constraint, applying the triangle inequality to the decomposition above and bounding the feedback term via the Rayleigh-Ritz theorem (as  $\mathbf{H}(\theta_i)$  has constant eigenvalues  $\pm \lambda_i$ ) yields  $\|\boldsymbol{\tau}_{\text{fb},i}\|_\infty \leq \lambda_i \|\mathbf{w}_i\|_2^2$ . Similarly, writing  $\xi_i$  in polar coordinates shows  $\|\boldsymbol{\tau}_{\text{ff},i}\|_\infty \leq c_i \sigma \|\xi_i\|_2$ , when  $\mathbf{h}_i = \mathbf{0}$  (Assumption 5). A sufficient condition for the torque constraint is therefore:

$$\|\xi_i\|_2 \leq \frac{1}{c_i \sigma} (\tau_{\max} - \lambda_i \|\mathbf{w}_i\|_2^2),$$

which requires the precondition  $\|\mathbf{w}_i\|_2^2 \leq \tau_{\max}/\lambda_i$ . For the wheel velocity constraint, the kinematics yield  $\|\dot{\Phi}_i\|_\infty \leq z_i \|\mathbf{w}_i\|_2$ , giving the sufficient condition  $\|\mathbf{w}_i\|_2 \leq \Phi_{\max}/z_i$ .

This defines the robust, heading-independent admissible set  $\mathcal{Z}_{\text{approx}}$ :

$$\mathcal{Z}_{\text{approx}} = \{(\eta_i, \xi_i) \mid \|\xi_i\|_2 \leq \frac{1}{c_i \sigma} (\tau_{\max} - \lambda \|\mathbf{w}_i\|_2^2), \|\mathbf{w}_i\|_2 \leq w_{\text{const},i}\}, \quad (14)$$

where  $w_{\text{const},i} := \min(\Phi_{\max}/z_i, \sqrt{\tau_{\max}/\lambda_i})$ .

*Remark 3:* Geometrically, the true set of feasible inputs  $\xi_i$  for a given orientation  $\theta_i$  is a parallelogram in input space (the pre-image of the torque box under the map  $\boldsymbol{\mu}(\theta_i)$ ). Our approach finds the largest inscribed circle across all orientations  $\theta_i \in \Theta$ , giving a single  $\theta$ -independent ball constraint scaled by  $c_i \sigma$ .

The terminal set  $\mathcal{X}_f$  is the subset of  $\mathcal{Z}_{\text{approx}}$  where the terminal law  $\kappa_f(\eta) = -\mathbf{K}\eta$  is admissible:

$$\mathcal{X}_f := \{\eta \in \mathbb{R}^4 \mid \|\mathbf{K}\eta\|_2 + \frac{\lambda_i}{c_i \sigma} \|\mathbf{S}\eta - \dot{\mathbf{h}}_i\|_2^2 \leq \frac{\tau_{\max}}{c_i \sigma} \text{ and } \|\mathbf{S}\eta - \dot{\mathbf{h}}_i\|_2^2 \leq w_{\text{const},i}^2\}, \quad (15)$$

where  $\mathbf{S} = [\mathbf{0}_{2 \times 2}, \mathbf{I}_2]$  extracts  $\dot{\mathbf{e}}_i$  from  $\eta_i$ . This set also defines the  $N$ -step region of attraction,  $\mathcal{X}_0$ , from which a feasible control sequence exists to steer the system into  $\mathcal{X}_f$ .

## V. COMPUTATIONALLY TRACTABLE NMPC DESIGN

Problem  $\mathcal{P}(i)$  remains intractable for real-time implementation due to the non-convexity of  $\mathcal{Z}_{\text{approx}}$ . We systematically

reformulate this into a computationally efficient SOCP. We substitute the path constraints  $(\boldsymbol{\eta}_{i,l|k}, \boldsymbol{\xi}_{i,l|k}) \in \mathcal{Z}_{\text{approx}}$  with five SOCP-compliant constraints using auxiliary variables  $[s_{i,k}, t_{i,k}]^\top \geq 0$ :

$$0 \leq s_{i,k} \leq w_{\text{const},i}^2, \quad 0 \leq t_{i,k} \leq w_{\text{const},i}, \quad (16a)$$

$$\|\mathbf{S}\boldsymbol{\eta}_{i,l|k} - \dot{\mathbf{h}}_{i,l|k}\|_2 \leq t_{i,k}, \quad (16b)$$

$$\left\| \begin{bmatrix} 2t_{i,k} \\ s_{i,k} - 1 \end{bmatrix} \right\|_2 \leq s_{i,k} + 1, \quad (16c)$$

$$\|\boldsymbol{\xi}_{i,l|k}\|_2 \leq \frac{1}{c_i\sigma}(\tau_{\max} - \lambda s_{i,k}), \quad (16d)$$

$$(16e)$$

**Equivalence:** The rotated second-order cone constraint (16c) is an equivalent conic form of the convex inequality  $t_{i,k}^2 \leq s_{i,k}$ : squaring both sides and performing some algebra reduces it to  $4t_{i,k}^2 \leq 4s_{i,k}$ . Combined with (16b) and (16a), this recovers exactly the quadratic form  $\|\mathbf{S}\boldsymbol{\eta}_{i,l|k} - \dot{\mathbf{h}}_{i,l|k}\|_2^2 \leq s_{i,k}$ , making it a lossless conic reformulation rather than a conservative relaxation.

To ensure tightness, we augment the objective function  $J_i$  to obtain:

$$J_{\text{final},i} = J_i + \rho_s(s_{i,k} - w_{\text{const},i}^2)^2 + \rho_t(t_{i,k} - w_{\text{const},i})^2, \quad (17)$$

where  $\rho_s, \rho_t$  are weighting parameters. The final SOCP problem  $\mathcal{P}_{\text{convex}}(i)$  to be solved by each robot  $i$  at time  $k$  is given by

$$\arg \min_{\boldsymbol{\Xi}_{i,k}, s_{i,k}, t_{i,k}} J_{\text{final},i}(\boldsymbol{\eta}_{i,0|k}, \boldsymbol{\Xi}_{i,k}, s_{i,k}, t_{i,k}) \quad (18a)$$

s.t.

$$\boldsymbol{\eta}_{i,l+1|k} = \hat{\mathbf{H}}\boldsymbol{\eta}_{i,l|k} + \hat{\mathbf{G}}\boldsymbol{\xi}_{i,l|k}, \quad \forall l \in [0, N-1], \quad (18b)$$

$$\tau_{\text{lb}} \leq \tau_i(\theta_{i,0|k}, \boldsymbol{\eta}_{i,0|k}, \boldsymbol{\xi}_{i,0|k}) \leq \tau_{\text{ub}} \quad (18c)$$

$$(16a)-(16d), \quad \forall l \in [1, N-1], \quad (18d)$$

$$(16a)-(16c), \quad l = N, \quad (18e)$$

$$\|\mathbf{K}\boldsymbol{\eta}_{i,N|k}\|_2 \leq \frac{1}{c_i\sigma}(\tau_{\max} - \lambda s_{i,k}). \quad (18f)$$

At step  $l = 0$ , we reduce conservatism by enforcing the exact physical constraint, Eq. (18c), which is a linear polytope based on (13), since the current heading  $\theta_{i,0|k}$  is known.

With the controller design and its convex reformulation established, we now state the main theoretical result of this paper.

**Theorem 1 (Closed-Loop Stability):** Suppose

Assumptions 1–6 hold and that the initial formation state of each agent satisfies  $\boldsymbol{\eta}_{i,0} \in \mathcal{X}_0$ , where  $\mathcal{X}_0$  denotes the  $N$ -step region of attraction of the closed-loop error dynamics. Define  $\mathbf{P}_i^{\text{ref}} = \mathbf{P}_{\text{vc}} + \boldsymbol{\Delta}_i$ . Let  $\mathcal{V}$  denote the set of all robots in the directed graph. For all  $t \geq 0$ , if the reference trajectories satisfy

$$\|\dot{\mathbf{P}}_i^{\text{ref}}(t)\|_2 < \frac{w_{\text{const},i}}{g_i}, \quad \|\ddot{\mathbf{P}}_i^{\text{ref}}(t)\|_2 < \frac{\tau_{\max} - \lambda_i w_{\text{const},i}^2}{c_i g_i \sigma}, \quad (19)$$

then for each robot  $i \in \mathcal{V}$ , the controller obtained by solving Problem  $\mathcal{P}_{\text{convex}}(i)$  implies the following for all  $t \geq 0$ :

- 1) recursive feasibility of the optimisation problem,
- 2) local asymptotic stability of the formation error,
- 3) local asymptotic stability of the overall nonlinear system

(2).

*Proof:* See Section VIII-B. ■

## VI. SIMULATION AND RESULTS

In this section, the proposed feedback linearised MPC, reformulated as an SOCP, is benchmarked against three alternatives: standard NMPC and LMPC controllers, and the NFLMPC controller.

### A. Baselines and Setup

We benchmark the proposed SOCP against NMPC (solves full nonlinear dynamics), LMPC (a Jacobian-based QP penalising torques directly), and NFLMPC (solves Problem  $\mathcal{P}(i)$  (12)). All simulations were performed on an Intel Core i7 CPU with 16 GB of RAM using MATLAB. LMPC (QP) was solved using qpOASES, and the proposed SOCP was solved using MOSEK 9.2 with default tolerances, while NMPC and NFLMPC were solved using IPOPT. All four controllers used the same prediction horizon ( $N = 10$ ) and sampling period ( $\delta_t = 0.1$  s).

**Robot Parameters:** The three differential-drive WMRs used in the experiments have mass  $m = 4.5$  kg, body-frame moment of inertia  $J_B = 0.057$  kg m<sup>2</sup>, wheel radius  $r = 0.05$  m, axle semi-length  $b = 0.125$  m, and centre-of-mass offset  $D = 0.11$  m. Actuator limits are  $\tau_{\max} = 0.25$  Nm and  $\dot{\Phi}_{\max} = 20.0$  rad/s. These parameters satisfy Assumption 4 by construction. Initial positions and headings of the virtual centre and the three WMRs are chosen arbitrarily.

**Scenarios and Protocol:** All scenarios share the same robot team, initial poses, start  $[-0.2, -1.5]^\top$  and goal  $[2.2, 4.8]^\top$ , and obstacles with a radius of 0.5 m, differing only in the number and placement of those obstacles relative to the virtual centre's path. Scenario 1 keeps three obstacles clear of the direct path to the goal; Scenarios 2 and 3 move them onto and around it; and Scenarios 4 and 5 add a fourth obstacle and tighten the cluster near the start, so obstacle density increases from 1 to 5. Each scenario was run 10 times, and the reported figures are averaged, which mainly stabilises the solver-time measurements.

**Metrics:** Torque variation is the total variation of the commanded wheel torques over a run,  $\sum_k \|\tau_i(k+1) - \tau_i(k)\|_1$ , summed over both wheels and averaged across the robots (Nm). As an accumulated sum of increments, it is not comparable to the per-step limit  $\tau_{\max} = 0.25$  Nm; smaller values mean smoother commands.

**Communication Topology and Formation Objective:**

The communication network between the virtual centre and the robots is defined by

$$\mathcal{A} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}, \quad \mathcal{B} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}. \quad (20)$$

The desired triangular formation is described by

$$\boldsymbol{\Delta} = \begin{bmatrix} 0.5 & -0.25 & -0.25 \\ 0 & -\frac{\sqrt{3}}{4} & \frac{\sqrt{3}}{4} \end{bmatrix}, \quad (21)$$

which the robots must maintain while tracking the motion of the virtual centre (See Fig. 2).

*Reference Generation via APF:* The virtual centre  $\mathbf{P}_{vc}(t)$  is driven by a modified Artificial Potential Field (APF) that addresses the Goals-Non-Reachable-with-Obstacles-Nearby (GNRON) problem. The attractive force is  $\mathbf{F}_{att} = k_{att}(\mathbf{P}_{goal} - \mathbf{P}_{vc})$ , and the repulsive force from the  $j$ -th obstacle is scaled by the squared goal distance  $d_{goal}^2$  to avoid local minima near the goal,

$$\mathbf{F}_{rep,j} = k_{rep} \left( \frac{1}{d_j} - \frac{1}{\rho_0} \right) \frac{d_{goal}^2}{d_j^3} \nabla d_j - k_{rep} \left( \frac{1}{d_j} - \frac{1}{\rho_0} \right)^2 (\mathbf{P}_{vc} - \mathbf{P}_{goal}), \quad (22)$$

with  $k_{att}, k_{rep} > 0$  and  $d_j$  the distance to obstacle  $j$ . Projecting the total field  $\mathbf{F}_{APF} = \mathbf{F}_{att} + \sum_j \mathbf{F}_{rep,j}$  into the body frame as  $[F_{||}, F_{\perp}]^T = \mathbf{R}(-\alpha_{vc}) \mathbf{F}_{APF}$  and setting  $[v_{vc}, \omega_{vc}] = [k_v F_{||}, k_{\omega} F_{\perp}]$  with  $k_v, k_{\omega} > 0$  gives the virtual-centre dynamics  $\dot{\mathbf{P}}_{vc} = \mathbf{R}(\alpha_{vc}) [v_{vc}, D\omega_{vc}]^T$  and  $\dot{\alpha}_{vc} = \omega_{vc}$ . As  $\mathbf{P}_{vc} \rightarrow \mathbf{P}_{goal}$ , the field  $\mathbf{F}_{APF} \rightarrow \mathbf{0}$ , so  $v_{vc}$  and  $\omega_{vc}$  vanish at the goal and the formation error converges to zero. The APF gains were tuned so that the path stays within the torque limits.

*Remark 4:* The APF method is not part of the proposed control scheme; it is used solely as a reference generator and may be replaced by any method that satisfies the admissibility assumptions.

The full code is available at <sup>1</sup>.

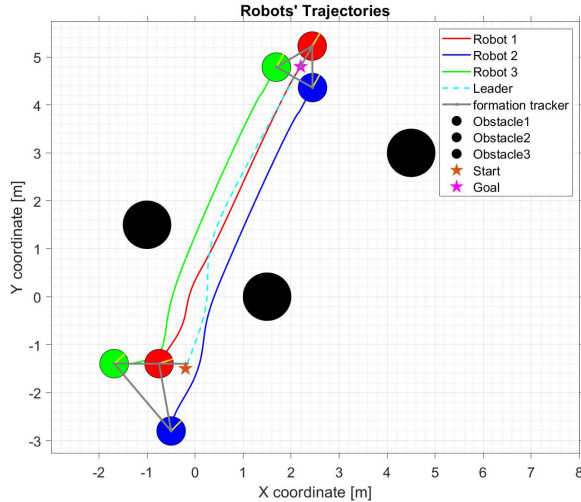


Fig. 2. Illustration of formation control of three robots in scenario 2 using the proposed SOCP controller.

## B. Results

Table I reports performance across the five scenarios. For a fair comparison, the state and input weighting matrices of LMPC, NMPC, and Nonconvex FLMPC were tuned until their RMS consensus errors agreed to within 1% ( $\approx 4.74$  m), isolating differences in computational load, torque smoothness, and settling behaviour. The proposed SOCP controller was not tuned to match this value due to conservatism arising from its input constraints.

Under matched tracking performance, NMPC yields the smallest torque variation (4.50 Nm) but requires 22.7 ms per solve. LMPC lies at the opposite extreme: extremely fast (0.049 ms) but with the largest torque variation (5.96 Nm). Nonconvex FLMPC provides a middle ground (4.69 Nm, 16.6 ms).

The proposed SOCP achieves similar torque smoothness (4.66 Nm) to Nonconvex FLMPC while reducing the average solve time to 2.24 ms, a  $7.4\times$  speed-up and more than an order-of-magnitude improvement over NMPC.

The SOCP exhibits a higher RMS error (5.787 m) due to conservatism from the wheel-speed limit (20 rad/s) and torque limit (0.25 Nm), which restricts the robot's linear velocity to roughly 0.66 m/s, whereas the other controllers can reach  $\approx 1$  m/s. Despite this reduced actuation authority, the SOCP achieves faster settling times than LMPC in three of the five scenarios.

To confirm that this gap is constraint-driven, all five scenarios were rerun with relaxed limits (0.5 Nm torque, 30 rad/s wheel speed) with the same weights. Its RMS error then fell to 4.32–4.77 m, matching the other controllers, at solver times of 2.3–2.5 ms, while retaining the smooth torque profile across the scenarios. Once actuation authority is no longer the bottleneck, the SOCP recovers the tracking performance of the Nonconvex FLMPC, confirming that the earlier gap stems from tight actuator limits rather than the formulation itself.

NMPC and Nonconvex FLMPC achieve nearly identical tracking and settling, with the latter solving faster (16.6 ms against 22.7 ms). All controllers satisfied the torque and wheel-velocity constraints. The reported timings were obtained in MATLAB with MOSEK on a desktop CPU, so deployment on a resource-constrained system remains to be verified.

## VII. CONCLUSION

In this work, we presented a distributed, torque-level formation control framework for nonholonomic wheeled mobile robots. By combining feedback linearisation with a consensus-based MPC architecture and reformulating the resulting non-convex optimal control problem as a tractable SOCP, we addressed the challenge of enforcing actuator-level constraints within a computationally efficient predictive control scheme. The proposed controller retains the smooth torque behaviour characteristic of feedback-linearised MPC while reducing average solver times from tens of milliseconds (NMPC) to approximately 2 ms in MATLAB. These timings indicate that the convex formulation is well-suited to real-time operation, although confirming this on a resource-constrained system remains to be done. The consensus structure further supports decentralised implementation and provides a promising path toward scalability.

Several directions remain open. Because the feedback linearisation exactly cancels the dynamics, its accuracy depends on well-identified inertial and geometric parameters; adaptive feedback-linearisation schemes that estimate uncertain inertial and geometric parameters [22], together with

<sup>1</sup><https://doi.org/10.5281/zenodo.20805007>

TABLE I  
CONTROLLER PERFORMANCE COMPARISON ACROSS FIVE SCENARIOS.

| Controller              | Metrics           | Scen. 1 | Scen. 2 | Scen. 3 | Scen. 4 | Scen. 5 | Average |
|-------------------------|-------------------|---------|---------|---------|---------|---------|---------|
| <b>LMPC</b>             | Solver Time [ms]  | 0.0556  | 0.0519  | 0.0458  | 0.0512  | 0.0413  | 0.0492  |
|                         | Settling Time [s] | 10      | 12      | 25      | 25      | 25      | 19.4    |
|                         | Torque Var. [Nm]  | 3.5954  | 4.0799  | 5.1752  | 6.3747  | 8.5546  | 5.956   |
|                         | RMS Error [m]     | 4.7971  | 4.7662  | 4.7776  | 4.7795  | 4.5903  | 4.742   |
| <b>NMPC</b>             | Solver Time [ms]  | 22.983  | 22.544  | 22.604  | 23.005  | 22.539  | 22.735  |
|                         | Settling Time [s] | 10      | 10      | 10      | 10      | 20      | 12      |
|                         | Torque Var. [Nm]  | 2.6084  | 3.2644  | 4.4882  | 5.2386  | 6.8779  | 4.4955  |
|                         | RMS Error [m]     | 4.7976  | 4.7649  | 4.7790  | 4.7719  | 4.5919  | 4.741   |
| <b>Nonconvex FL MPC</b> | Solver Time [ms]  | 16.765  | 16.666  | 16.750  | 16.863  | 16.168  | 16.642  |
|                         | Settling Time [s] | 10      | 10      | 10      | 10      | 20      | 12      |
|                         | Torque Var. [Nm]  | 2.6423  | 3.2635  | 4.3317  | 5.8029  | 7.4309  | 4.694   |
|                         | RMS Error [m]     | 4.7969  | 4.7623  | 4.7768  | 4.7731  | 4.5954  | 4.741   |
| <b>Proposed (SOCP)</b>  | Solver Time [ms]  | 2.2814  | 2.3014  | 2.3017  | 2.1697  | 2.1440  | 2.240   |
|                         | Settling Time [s] | 15      | 15      | 15      | 15      | 20      | 16      |
|                         | Torque Var. [Nm]  | 2.4273  | 3.2922  | 4.5607  | 5.8033  | 7.2196  | 4.661   |
|                         | RMS Error [m]     | 6.1517  | 5.9807  | 5.8066  | 6.0718  | 4.9232  | 5.787   |

approximation-based extensions for unmodelled dynamics, offer a route to relaxing this dependence. We also intend to relax the constant-velocity neighbour approximation of Assumption 5, for instance by exchanging predicted trajectories between agents, and to address communication delays, packet loss, and imperfect state information. As the terminal constraint set can be conservative, reshaping it to enlarge the region of attraction and improve robustness is a further direction. Finally, embedding CBFs within the convex MPC would guarantee inter-agent safety at little cost, and validation on physical platforms will help bridge theory and practice.

## VIII. APPENDIX

### A. Proof of Lemma 1

From the dynamics and the two-stage linearisation,  $\mathbf{g}_e(\mathbf{x}_i)$  can be defined by the following expression:

$$\mathbf{g}_e(\mathbf{x}_i) = g_i \frac{r}{2b} \mathbf{R}(\theta_i) \begin{bmatrix} b & b \\ D & -D \end{bmatrix}.$$

$\mathbf{R}(\theta_i) \in \text{SO}(2)$ , thus the determinant of the decoupling matrix can be expressed as  $\det(\mathbf{g}_e(\mathbf{x}_i)) = -\frac{r^2 g_i^2 D}{2b}$ . Under Assumptions 1 and 4,  $\det(\mathbf{g}_e(\mathbf{x}_i)) \neq 0$ ,  $\forall \mathbf{x}_i \in \mathbb{X}$ , therefore  $\mathbf{g}_e(\mathbf{x}_i)$  is nonsingular, and as a result, the feedback linearisation always remains valid. ■

### B. Proof of Theorem 1

1) *Recursive Feasibility & Local Asymptotic Stability of Formation Error*: The proof follows the standard Lyapunov arguments for constrained MPC in [21]. The admissibility conditions in (19) ensure that the reference velocities and accelerations remain within the inner-approximation set  $\mathcal{Z}_{\text{approx}}$ , so the state and input constraints induced by the torque and wheel-speed limits are compatible with the predicted closed-loop behaviour. Consequently, the terminal controller  $\kappa_f$ , terminal set  $\mathcal{X}_f$ , and terminal cost  $\ell_f$  satisfy the terminal admissibility, invariance, and cost-decrease properties described in Section III-B. Under these conditions,

the standard shift argument applies: letting  $V_i(\boldsymbol{\eta}_{i,k})$  denote the optimal cost at time  $k$ , the principle of optimality yields:  $V_i(\boldsymbol{\eta}_{i,k+1}) - V_i(\boldsymbol{\eta}_{i,k}) \leq -\ell_s(\boldsymbol{\eta}_{i,k}, \boldsymbol{\xi}_{i,0|k}^*)$ , for all  $k \in \mathbb{Z}^+$ , where  $\boldsymbol{\xi}_{i,0|k}^*$  is the first optimal input. Since  $\ell_s$  is positive definite,  $V_i$  is a discrete-time Lyapunov function for the closed-loop error dynamics, establishing recursive feasibility and local asymptotic stability of  $\boldsymbol{\eta}_{i,k}$ .

2) *Local Asymptotic Stability of Nonlinear System*: Given that the feedback linearisation is globally valid by construction (Lemma 1), it remains to show that the internal dynamics are stabilised under the closed-loop control law [15]. We identify the wheel velocity error as the internal state and show that it converges to zero as a direct consequence of the formation error convergence established in Section VIII-B.1.

a) *Convergence of the formation error*: From Section VIII-B.1,  $\boldsymbol{\eta}_{i,k} \rightarrow \mathbf{0}$  as  $k \rightarrow \infty$  for all  $i \in \mathcal{V}$ . Since  $\boldsymbol{\eta}_{i,k} = [\mathbf{e}_{i,k}^\top \quad \dot{\mathbf{e}}_{i,k}^\top]^\top$  by definition, then for all  $i \in \mathcal{V}$ ,

$$[\mathbf{e}_i(t)^\top, \dot{\mathbf{e}}_i(t)^\top]^\top \rightarrow \mathbf{0} \quad \text{as } t \rightarrow \infty, \quad (23)$$

where the passage from discrete-time to continuous-time convergence follows from Assumption 6.

b) *Convergence of the Cartesian velocity*: Define the global reference  $\mathbf{P}^{\text{ref}} = \mathbf{P}_{\text{vc}} \mathbf{1}_n^\top + \boldsymbol{\Delta}$ , stacking  $\mathbf{P}_i^{\text{ref}} = \mathbf{P}_{\text{vc}} + \boldsymbol{\Delta}_i$ . Since  $(\mathbf{P}^{\text{ref}} - \boldsymbol{\Delta})\mathcal{H} = \mathbf{P}_{\text{vc}} \mathbf{1}_n^\top \mathcal{H}$  and  $\mathbf{1}_n^\top \mathcal{L} = \mathbf{0}^\top$  (Assumption 1) yields  $\mathbf{1}_n^\top \mathcal{H} = \mathbf{1}_n^\top \mathcal{B}$ , expanding Eq. (5) collapses to the identity  $\mathbf{e} = (\mathbf{P} - \mathbf{P}^{\text{ref}})\mathcal{H}$ . As  $\mathcal{H}$  is constant (Assumption 2) and nonsingular (Assumption 1), right multiplication by it is bijective, so  $\mathbf{P} - \mathbf{P}^{\text{ref}} = \mathbf{e}\mathcal{H}^{-1}$  and  $\mathbf{e}(t) \rightarrow \mathbf{0} \iff \mathbf{P}(t) \rightarrow \mathbf{P}^{\text{ref}}(t)$ . Under Assumption 3,  $\mathbf{P}^{\text{ref}}$  is differentiable, and differentiating gives  $\dot{\mathbf{e}} = (\dot{\mathbf{P}} - \dot{\mathbf{P}}^{\text{ref}})\mathcal{H}$ , so likewise  $\dot{\mathbf{e}}(t) \rightarrow \mathbf{0} \iff \mathbf{P}(t) \rightarrow \dot{\mathbf{P}}^{\text{ref}}(t)$ . Hence, for all  $i \in \mathcal{V}$ ,  $[\mathbf{e}_i^\top, \dot{\mathbf{e}}_i^\top]^\top \rightarrow \mathbf{0} \iff [\mathbf{P}_i^\top, \dot{\mathbf{P}}_i^\top]^\top \rightarrow [(\mathbf{P}_i^{\text{ref}})^\top, (\dot{\mathbf{P}}_i^{\text{ref}})^\top]^\top$  as  $t \rightarrow \infty$ .

c) *Internal state definition*: The robot kinematics relate  $\dot{\mathbf{P}}_i$  to the wheel velocities  $\dot{\boldsymbol{\Phi}}_i$  via:

$$\dot{\mathbf{P}}_i = \mathbf{S}_P(q_i) \dot{\boldsymbol{\Phi}}_i, \quad \mathbf{S}_P(q_i) = \frac{r}{2b} \mathbf{R}(\theta_i) \begin{bmatrix} b & b \\ D & -D \end{bmatrix}, \quad (24)$$

which can be inverted to give:

$$\begin{aligned}\dot{\Phi}_i &= \mathbf{S}_P(q_i)^{-1} \dot{\mathbf{P}}_i \\ &= \frac{1}{rD} \begin{bmatrix} D & b \\ D & -b \end{bmatrix} \mathbf{R}(\theta_i)^\top \dot{\mathbf{P}}_i \\ &\triangleq \mathbf{TR}(\theta_i)^\top \dot{\mathbf{P}}_i,\end{aligned}\quad (25)$$

where  $\mathbf{T} \triangleq \frac{1}{rD} \begin{bmatrix} D & b \\ D & -b \end{bmatrix}$  is nonsingular by Assumption 4, since  $\det(\mathbf{T}) = -2b/(rD) \neq 0$ . We define the reference wheel velocity using the current heading  $\theta_i$  consistently with (25):

$$\dot{\Phi}_i^{\text{ref}} \triangleq \mathbf{TR}(\theta_i)^\top \dot{\mathbf{P}}_i^{\text{ref}}, \quad (26)$$

and define the internal state as the wheel velocity error:

$$\mathbf{z}_i \triangleq \dot{\Phi}_i - \dot{\Phi}_i^{\text{ref}} = \mathbf{TR}(\theta_i)^\top (\dot{\mathbf{P}}_i - \dot{\mathbf{P}}_i^{\text{ref}}), \quad (27)$$

where  $\mathbf{z}_i$  is a linear function of the Cartesian velocity error via the composite map  $\mathbf{TR}(\theta_i)^\top$ .

d) *Convergence of the internal state*: Taking the induced 2-norm of (27) and applying submultiplicativity of the induced 2-norm gives

$$\|\mathbf{z}_i\|_2 \leq \|\mathbf{T}\|_2 \|\mathbf{R}(\theta_i)^\top\|_2 \|\dot{\mathbf{P}}_i - \dot{\mathbf{P}}_i^{\text{ref}}\|_2. \quad (28)$$

Since  $\mathbf{R}(\theta_i) \in SO(2)$ , the factor  $\|\mathbf{R}(\theta_i)^\top\|_2 = 1$  holds uniformly for all  $\theta_i \in \Theta$ . The factor  $\|\mathbf{T}\|_2 = \sigma_{\max}(\mathbf{T})$  is a finite positive constant determined solely by the robot geometry  $(r, b, D)$  under Assumption 4. Hence:

$$\|\mathbf{z}_i\|_2 \leq \sigma_{\max}(\mathbf{T}) \|\dot{\mathbf{P}}_i - \dot{\mathbf{P}}_i^{\text{ref}}\|_2.$$

Since as  $t \rightarrow \infty$ ,  $\|\dot{\mathbf{P}}_i - \dot{\mathbf{P}}_i^{\text{ref}}\|_2 \rightarrow 0$ , and  $\sigma_{\max}(\mathbf{T})$  is independent of  $\theta_i(t)$ , it follows that:

$$\mathbf{z}_i(t) \rightarrow \mathbf{0} \quad \text{as } t \rightarrow \infty. \quad (29)$$

This conclusion holds for all  $\theta_i(t) \in \Theta$  and for all reference trajectories satisfying Assumption 3.

e) *Local asymptotic stability of the full nonlinear system*: From Steps a)-d), the controlled states satisfy:

$$\boldsymbol{\eta}_i(t) \rightarrow \mathbf{0} \quad \text{and} \quad \mathbf{z}_i(t) \rightarrow \mathbf{0} \quad \text{as } t \rightarrow \infty.$$

It remains to characterise the behaviour of the heading  $\theta_i$ . From the kinematic relation:

$$\dot{\theta}_i = \frac{r}{2b} [1, -1] \dot{\Phi}_i, \quad (30)$$

and the wheel velocity convergence (29), it follows that:

$$\dot{\theta}_i(t) \rightarrow \frac{r}{2b} [1, -1] \dot{\Phi}_i^{\text{ref}}(t) \quad \text{as } t \rightarrow \infty. \quad (31)$$

That is, the heading rate asymptotically tracks the reference heading rate induced by  $\dot{\mathbf{P}}_i^{\text{ref}}$ . Furthermore, since  $\theta_i \in \Theta = (-\pi, \pi]$  by definition of the configuration space, the heading remains bounded for all  $t \geq 0$ .

Since the feedback linearisation is globally valid (Lemma 1), and as  $t \rightarrow \infty$ ,  $\boldsymbol{\eta}_i \rightarrow \mathbf{0}$ ,  $\mathbf{z}_i \rightarrow \mathbf{0}$ , and  $\theta_i$  remains bounded in  $\Theta$  with  $\theta_i$  asymptotically tracking the reference heading rate, the full state  $\mathbf{x}_i = (q_i, \dot{\Phi}_i)$  of the nonlinear system (2) is locally asymptotically stable about the reference trajectory for all  $i \in \mathcal{V}$  [6]. ■

## REFERENCES

- [1] P. R. Wurman, R. D'Andrea, and M. Mountz, "Coordinating Hundreds of Cooperative, Autonomous Vehicles in Warehouses," *AI Magazine*, vol. 29, no. 1, pp. 9–19, 2008.
- [2] R. Beard, T. McLain, D. Nelson, D. Kingston, and D. Johanson, "Decentralized Cooperative Aerial Surveillance Using Fixed-Wing Miniature UAVs," *Proceedings of the IEEE*, vol. 94, no. 7, pp. 1306–1324, Jul. 2006, conference Name: Proceedings of the IEEE.
- [3] J. Zhang, M. Liu, S. Zhang, R. Zheng, and S. Dong, "Multi-AUV Adaptive Path Planning and Cooperative Sampling for Ocean Scalar Field Estimation," *IEEE Transactions on Instrumentation and Measurement*, vol. 71, pp. 1–14, 2022, conference Name: IEEE Transactions on Instrumentation and Measurement.
- [4] K.-K. Oh, M.-C. Park, and H.-S. Ahn, "A survey of multi-agent formation control," *Automatica*, vol. 53, pp. 424–440, Mar. 2015.
- [5] R. J. Stern, "Brockett's stabilization condition under state constraints," *Systems & Control Letters*, vol. 47, no. 4, pp. 335–341, Nov. 2002.
- [6] H. K. Khalil, *Nonlinear Systems*. Prentice Hall, 2002.
- [7] Y. Cao, W. Yu, W. Ren, and G. Chen, "An Overview of Recent Progress in the Study of Distributed Multi-Agent Coordination," *IEEE Transactions on Industrial Informatics*, vol. 9, no. 1, pp. 427–438, Feb. 2013, conference Name: IEEE Transactions on Industrial Informatics.
- [8] M. A. Lewis and K.-H. Tan, "High Precision Formation Control of Mobile Robots Using Virtual Structures," *Autonomous Robots*, vol. 4, no. 4, pp. 387–403, Oct. 1997.
- [9] T. Balch and R. Arkin, "Behavior-based formation control for multi-robot teams," *IEEE Transactions on Robotics and Automation*, vol. 14, no. 6, pp. 926–939, Dec. 1998, conference Name: IEEE Transactions on Robotics and Automation.
- [10] K. Li, Y. Bian, S. E. Li, B. Xu, and J. Wang, "Distributed model predictive control of multi-vehicle systems with switching communication topologies," *Transportation Research Part C: Emerging Technologies*, vol. 118, p. 102717, Sep. 2020.
- [11] X. Zhang, W. Pan, C. Li, X. Xu, X. Wang, R. Zhang, and D. Hu, "Toward Scalable Multirobot Control: Fast Policy Learning in Distributed MPC," *IEEE Transactions on Robotics*, vol. 41, pp. 1491–1512, 2025, conference Name: IEEE Transactions on Robotics.
- [12] N. Nfaileh, K. Alipour, B. Tarvirdzadeh, and A. Hadi, "Formation control of multiple wheeled mobile robots based on model predictive control," *Robotica*, vol. 40, no. 9, pp. 3178–3213, Sep. 2022.
- [13] R. Fierro and F. L. Lewis, "Control of a nonholonomic mobile robot: Backstepping kinematics into dynamics," *Journal of Robotic Systems*, vol. 14, no. 3, pp. 149–163, 1997.
- [14] R. Olfati-Saber, J. A. Fax, and R. M. Murray, "Consensus and Cooperation in Networked Multi-Agent Systems," *Proceedings of the IEEE*, vol. 95, no. 1, pp. 215–233, Jan. 2007, conference Name: Proceedings of the IEEE.
- [15] S. Sastry, *Nonlinear Systems*, ser. Interdisciplinary Applied Mathematics, J. E. Marsden, L. Sirovich, and S. Wiggins, Eds. New York, NY: Springer, 1999, vol. 10.
- [16] C. Tiriolo and W. Lucia, "A Feedback Linearized Model Predictive Control Strategy for Input-Constrained Self-Driving Cars," May 2024, arXiv:2405.01753 [eess].
- [17] D. Simon, J. Löfberg, and T. Glad, "Nonlinear model predictive control using Feedback Linearization and local inner convex constraint approximations," in *2013 European Control Conference (ECC)*, Jul. 2013, pp. 2056–2061.
- [18] M. A. Kamel and Y. Zhang, "Linear model predictive control via feedback linearization for formation control of multiple wheeled mobile robots," in *2015 IEEE International Conference on Information and Automation*, Aug. 2015, pp. 1283–1288.
- [19] R. Siegwart and I. R. Nourbakhsh, *Introduction to Autonomous Mobile Robots*. MIT Press, 2004, google-Books-ID: gUbQ9.weg88C.
- [20] D. Nešić, A. R. Teel, and P. V. Kokotović, "Sufficient conditions for stabilization of sampled-data nonlinear systems via discrete-time approximations," *Systems & Control Letters*, vol. 38, no. 4, pp. 259–270, Dec. 1999.
- [21] D. Q. Mayne, J. B. Rawlings, C. V. Rao, and P. O. M. Scokaert, "Constrained model predictive control: Stability and optimality," *Automatica*, vol. 36, no. 6, pp. 789–814, Jun. 2000.
- [22] T. Fukao, H. Nakagawa, and N. Adachi, "Adaptive tracking control of a nonholonomic mobile robot," *IEEE Transactions on Robotics and Automation*, vol. 16, no. 5, pp. 609–615, 2000.